

A Symbolic Infinitesimal Flex for the Dimension-Seven Slice of a Volume-Rigidity Conjecture

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Abstract

Cruickshank, Jackson, and Tanigawa conjectured that if $\mathcal{S}$ is a simplicial 2-sphere on $\lceil d/2 \rceil + 2$ vertices and Z is a disjoint set of $d - 3$ vertices, then the hypergraph of $(d - 2)$ -faces of the simplicial cone $\mathcal{S} * Z$ is volume rigid in $\mathbb{R}^d$. We give a symbolic counterexample to the complete $d = 7$ slice. In fact, the same construction applies to every simplicial 2-sphere on six vertices. For a generic placement, form the 7×3 matrix D of the three differences among the four cone vertices. We construct a nonzero symmetric matrix $B = DKD^\top$ and the velocity field $u(v) = B(p(v) - p(z_0))$. Two trace identities, one for hyperedges of the form $e \cup Z$ and one for hyperedges of the form $t \cup (Z \setminus \{z\})$, show symbolically that all 44 squared five-volume derivatives vanish. The motion is not Euclidean-trivial. Thus the generic Jacobian rank is at most 41, below the rigidity target 42. Finite enumeration and exact computations are reported only as independent cross-checks of the combinatorics and the symbolic certificate.

1 Introduction

Cruickshank, Jackson, and Tanigawa introduced a framework for volume rigidity of uniform hypergraphs and simplicial complexes [CJT26]. Among their proposed base cases for higher-dimensional results is the following statement [CJT26, Conjecture 22]: for $d \geq 4$, if $\mathcal{S}$ is a simplicial 2-sphere on $\lceil d/2 \rceil + 2$ vertices and Z is a disjoint set of $d - 3$ vertices, then the hypergraph $H_{d-2}(\mathcal{S} * Z)$ is volume rigid in $\mathbb{R}^d$.

This note concerns exactly the slice $d = 7$. Here $\mathcal{S}$ has six vertices, Z has four vertices, and $H_5(\mathcal{S} * Z)$ has ten vertices and 44 hyperedges. A generic placement has 70 scalar velocity coordinates. The space of translations and infinitesimal rotations has dimension

$$7 + \binom{7}{2} = 28,$$

so infinitesimal volume rigidity would require rank $70 - 28 = 42$.

Our main result is a symbolic rank obstruction that is uniform over all six-vertex simplicial 2-spheres. It does not rely on a special placement, a finite-field rank computation, or the finite enumeration of the two combinatorial types. Those computations are included only as reproducibility checks. The result does not address $d \geq 8$, the broader manifold conjecture in [CJT26], nongeneric placements, or global uniqueness.

2 Definitions and the dimension-seven slice

2.1 Simplicial cones and their hypergraphs

For a simplicial complex $\mathcal{S}$ and a finite set Z disjoint from $V(\mathcal{S})$, define

$$\mathcal{S} * Z = \{X \cup Z' : X \in \mathcal{S}, Z' \subseteq Z\}.$$

For a simplicial complex $\mathcal{K}$, the hypergraph $H_k(\mathcal{K})$ has vertex set $V(\mathcal{K})$ and one hyperedge for every k -face of $\mathcal{K}$; hence every hyperedge has $k + 1$ vertices.

Fix a simplicial 2-sphere $\mathcal{S}$ on vertices $0, 1, \dots, 5$, and let

$$Z = \{6, 7, 8, 9\} = \{z_0, z_1, z_2, z_3\}, \quad z_i = 6 + i.$$

The six-element hyperedges of $H_5(\mathcal{S} * Z)$ are exactly

$$e \cup Z \quad (e \in \mathcal{S}^{(1)}), \quad t \cup (Z \setminus \{z_i\}) \quad (t \in \mathcal{S}^{(2)}, \ 0 \leq i \leq 3). \quad (1)$$

Indeed, a six-element face of the join contains either two sphere vertices and all four cone vertices, or three sphere vertices and three cone vertices.

Every triangulated 2-sphere on six vertices has twelve edges and eight triangles: Euler's relation and double counting edge-triangle incidences give

$$6 - f_1 + f_2 = 2, \quad 2f_1 = 3f_2,$$

and hence $(f_1, f_2) = (12, 8)$. Thus

$$|E(H_5(\mathcal{S} * Z))| = 12 + 4 \cdot 8 = 44.$$

2.2 The two six-vertex sphere types

For completeness, the two combinatorial types of six-vertex simplicial 2-sphere may be represented by the following triangle sets:

$$\begin{aligned} T_{\text{st}} &= \{012, 013, 023, 124, 134, 235, 245, 345\}, \\ T_{\text{oct}} &= \{012, 013, 024, 034, 125, 135, 245, 345\}. \end{aligned}$$

Their edge sets are, respectively,

$$\begin{aligned} E_{\text{st}} &= \{01, 02, 03, 12, 13, 14, 23, 24, 25, 34, 35, 45\}, \\ E_{\text{oct}} &= \{01, 02, 03, 04, 12, 13, 15, 24, 25, 34, 35, 45\}. \end{aligned}$$

The complete ordered lists of all 44 hyperedges for each type are given in Section A. The symbolic argument below only uses the two forms in (1), so it applies uniformly and does not depend on the classification.

2.3 Squared-volume Jacobian

Let $p : V(H_5(\mathcal{S} * Z)) \rightarrow \mathbb{R}^7$ be a placement. It is *generic* if its 70 scalar coordinates are algebraically independent over $\mathbb{Q}$. For a hyperedge $\Delta = \{v_0, v_1, \dots, v_5\}$, let $M_\Delta(p)$ be the 5×7 matrix whose i -th row is $(p(v_i) - p(v_0))^\top$, and put

$$g_\Delta(p) = \det(M_\Delta(p)M_\Delta(p)^\top).$$

Then

$$g_\Delta(p) = (5!)^2 \text{Vol}_5(p(\Delta))^2.$$

Let $J_\mathcal{S}(p)$ be the 44×70 Jacobian with one row $\nabla g_\Delta(p)$ for each hyperedge Δ . At a generic placement this Jacobian has the same rank and right kernel as the volume-rigidity matrix of [CJT26]: the Gram determinant differs from squared five-volume by the fixed nonzero factor $(5!)^2$, and the additional row scalings used in that paper are nonzero at a generic placement.

The Euclidean-trivial infinitesimal motions are

$$u(v) = Ap(v) + t, \quad A^\top = -A, \quad t \in \mathbb{R}^7.$$

A generic ten-point placement affinely spans $\mathbb{R}^7$, so these motions form a 28-dimensional subspace of $\ker J_\mathcal{S}(p)$. Consequently the $d = 7$ specialization of Conjecture 22 predicts $\text{rank } J_\mathcal{S}(p) = 42$.

3 Main result

Theorem 3.1. *Let $\mathcal{S}$ be any simplicial 2-sphere on six vertices, let Z be a disjoint four-element set, and let $p : V(H_5(\mathcal{S} * Z)) \rightarrow \mathbb{R}^7$ be generic. Then*

$$\text{rank } J_{\mathcal{S}}(p) \leq 41.$$

*In particular, $H_5(\mathcal{S} * Z)$ is not volume rigid in $\mathbb{R}^7$, and the $d = 7$ slice of Conjecture 22 in [CJT26] is false.*

The remainder of this section constructs a single additional symbolic infinitesimal motion. The equivalence between generic volume rigidity and generic infinitesimal volume rigidity is Proposition 6 of [CJT26].

3.1 A trace formula for squared volume

For a linear subspace $U \subseteq \mathbb{R}^7$, write P_U for its orthogonal projector.

Lemma 3.2. *Let Δ be a nondegenerate 5-simplex with direction space U , and let B be a symmetric 7×7 matrix. Under the affine velocity field $u(q) = B(q - q_0)$, where q_0 is fixed,*

$$\left. \frac{d}{dt} \right|_{t=0} g_{\Delta}(q_0 + (I + tB)(p - q_0)) = 2g_{\Delta}(p) \text{tr}(P_U B).$$

The same formula holds with g_{Δ} replaced by squared five-volume.

Proof. Translations do not affect edge differences. If $M = M_{\Delta}(p)$, then the transformed difference matrix is $M_t = M(I + tB)^{\top}$. Since $B = B^{\top}$,

$$\left. \frac{d}{dt} \right|_{t=0} M_t M_t^{\top} = 2MBM^{\top}.$$

Jacobi's determinant identity gives

$$\begin{aligned} \left. \frac{d}{dt} \right|_{t=0} \det(M_t M_t^{\top}) &= 2 \det(MM^{\top}) \text{tr}\left((MM^{\top})^{-1}MBM^{\top}\right) \\ &= 2g_{\Delta}(p) \text{tr}\left(M^{\top}(MM^{\top})^{-1}MB\right). \end{aligned}$$

The matrix $M^{\top}(MM^{\top})^{-1}M$ is P_U , which proves (3.2). Division by $(5!)^2$ gives the squared-volume version. $\square$

3.2 The symbolic motion

Translate the notation so that $p(z_0)$ is the origin, and define

$$D = \begin{bmatrix} d_1 & d_2 & d_3 \end{bmatrix} = \begin{bmatrix} p(z_1) - p(z_0) & p(z_2) - p(z_0) & p(z_3) - p(z_0) \end{bmatrix}, \quad G = D^{\top}D.$$

Write

$$a = G_{12}, \quad b = G_{13}, \quad c = G_{23},$$

and set

$$\xi = c - b, \quad \eta = a - c, \quad \zeta = b - a, \quad K = \begin{pmatrix} 0 & \xi & \eta \\ \xi & 0 & \zeta \\ \eta & \zeta & 0 \end{pmatrix}.$$

Finally, put

$$B = DKD^{\top}, \quad u(v) = B(p(v) - p(z_0)). \tag{2}$$

This is a polynomial velocity vector in the placement coordinates.

The definitions immediately give

$$\xi + \eta + \zeta = 0, \quad a\xi + b\eta + c\zeta = 0. \quad (3)$$

Consequently

$$\text{tr}(KG) = 0, \quad h^\top Kh = 0 \quad \text{for } h \in \{e_1, e_2, e_3, (1, 1, 1)^\top\}. \quad (4)$$

The first equation follows because $\text{tr}(KG) = 2(a\xi + b\eta + c\zeta)$. The three coordinate-vector equations in the second statement are the zero diagonal of K , and the remaining equation is $2(\xi + \eta + \zeta) = 0$.

The four covectors in (4) are normal covectors to the four facets of the tetrahedron with coordinates $0, e_1, e_2, e_3$ in the basis D . For a generic placement, D has full column rank and a, b, c are not all equal. The latter equalities define a proper algebraic subset. Hence $K \neq 0$, and full column rank of D then implies $B = DKD^\top \neq 0$. The matrix B is symmetric.

3.3 Stationarity of all hyperedge volumes

Let $L = \text{im } D$. The operator B is supported on L : it kills $L^\perp$, maps into L , and satisfies

$$\text{tr}(B) = \text{tr}(DKD^\top) = \text{tr}(KG) = 0. \quad (5)$$

First consider a hyperedge $\Delta = e \cup Z$. Its direction space has the orthogonal decomposition

$$U_\Delta = L \oplus R,$$

where $R \subseteq L^\perp$ is spanned by the projections of the two sphere vertices in e . Therefore

$$\text{tr}(P_{U_\Delta} B) = \text{tr}(P_L B) + \text{tr}(P_R B) = \text{tr}(B) = 0. \quad (6)$$

Now let $\Delta = t \cup (Z \setminus \{z_\ell\})$. Denote by L_ℓ the two-dimensional direction space of the corresponding facet of the tetrahedron on Z . Let h_ℓ be its normal covector in D -coordinates; thus

$$\{h_0, h_1, h_2, h_3\} = \{(1, 1, 1)^\top, e_1, e_2, e_3\}.$$

The vector

$$n_\ell = DG^{-1}h_\ell$$

is normal to L_ℓ inside L , because $D^\top n_\ell = h_\ell$. From (4),

$$n_\ell^\top B n_\ell = h_\ell^\top K h_\ell = 0. \quad (7)$$

Since $L = L_\ell \oplus \text{span}(n_\ell)$ orthogonally, (5) and (7) imply

$$\text{tr}(P_{L_\ell} B) = 0.$$

The direction space of Δ decomposes as

$$U_\Delta = L_\ell \oplus R_\ell$$

for a three-dimensional subspace $R_\ell \subseteq L_\ell^\perp$. Moreover,

$$L_\ell^\perp = \text{span}(n_\ell) \oplus L^\perp.$$

For $w = \lambda n_\ell + r$, with $r \in L^\perp$, the facts $Br = 0$ and (7) give $w^\top B w = 0$. Because B is symmetric, polarization shows that its compression to $L_\ell^\perp$ is zero. Hence $\text{tr}(P_{R_\ell} B) = 0$, and altogether

$$\text{tr}(P_{U_\Delta} B) = 0. \quad (8)$$

Equations (6) and (8) cover the two families in (1), hence all 44 hyperedges. By Lemma 3.2, the velocity u in (2) lies in $\ker J_S(p)$.

3.4 Nontriviality and the rank bound

Suppose, for contradiction, that u is Euclidean-trivial. Then there are a skew-symmetric matrix A and a vector t such that

$$B(p(v) - p(z_0)) = Ap(v) + t$$

for all ten vertices. Substitution of $v = z_0$ gives $t = -Ap(z_0)$, so

$$B(p(v) - p(z_0)) = A(p(v) - p(z_0))$$

for every vertex v . A generic ten-point placement affinely spans $\mathbb{R}^7$; its differences from $p(z_0)$ therefore span $\mathbb{R}^7$, forcing $B = A$. This is impossible because B is nonzero and symmetric while A is skew-symmetric.

Thus $\ker J_S(p)$ contains the 28-dimensional trivial motion space and the independent motion u . Rank-nullity yields

$$\text{rank } J_S(p) \leq 70 - (28 + 1) = 41,$$

which proves Theorem 3.1.

4 Exact cross-checks and verification boundary

The proof of Theorem 3.1 is the symbolic argument in the preceding section. The following finite computations do not carry the generic conclusion.

An exhaustive enumeration considered all $\binom{20}{8} = 125970$ choices of eight triangles on six labelled vertices. A topology filter retained 195 labelled sphere face tables. Quotienting by the S_6 -action gave the two types in Section 2.2, with labelled orbit sizes 180 and 15, and automorphism-group orders 4 and 48, respectively. An independent graph-atlas reconstruction found the same two planar, three-connected, six-vertex triangulations.

At one exact integer placement (deterministic seed 260830), each type passed the following checks: the ten points had affine rank seven; all 44 Gram determinants were nonzero; the 28 standard trivial motions and the motion in (2) lay in the exact kernel; and adding the displayed motion raised the motion-space rank from 28 to 29. The specialized Jacobian had rank 40 modulo the prime 2147483647. Exact 40×40 minors were nonzero, with 394 and 395 decimal digits for the stacked and octahedral types. These specialized calculations are cross-checks and lower-rank witnesses at the chosen coordinates; they are not the source of the generic upper bound.

A separate checker reconstructed the two sphere types from the graph atlas, rebuilt all 44 rows using Cayley–Menger determinant derivatives, verified the identities in (3)–(4), and recomputed the exact minors with NetworkX 3.5 and SymPy 1.14.0. It reported **PASS**. That checker reused the stored face tables, coordinates, and candidate minor indices, so it was not completely input-isolated; its independent row construction and symbolic checks nevertheless provide a noncircular cross-check. The generic claim remains the symbolic proof above.

The mathematical argument received an independent verdict of **CORRECT**. A separate paper-level review checked the theorem quantifiers, the expanded determinant calculation in Lemma 3.2, the source interfaces, and the complete hyperedge tables, and likewise reported **CORRECT**.

5 Relation to the source paper and scope

The source paper proves volume-rigidity results for lower skeletons and for the codimension-two manifold case in dimensions 4, 5, 6, and it states Conjecture 22 as a proposed family of base cases for extending the method [CJT26]. The theorem here addresses only the next numerical slice, $d = 7$, and only the simplicial-cone hypergraphs appearing in that conjecture. It does not

settle the source paper’s broader Conjecture 2, since that extension requires further arguments even if all the proposed base cases were available.

This note makes no claim of public priority, of being the first counterexample, or of establishing the current public status of the all-dimensional conjecture. Public novelty or priority is NOT ESTABLISHED.

A Complete hyperedge lists

In the tables below, a string such as 012678 denotes the six-element set $\{0, 1, 2, 6, 7, 8\}$. Each table contains all 44 hyperedges in the order used for the exact cross-check.

Stacked type

012678	012679	012689	012789
013678	013679	013689	013789
016789	023678	023679	023689
023789	026789	036789	124678
124679	124689	124789	126789
134678	134679	134689	134789
136789	146789	235678	235679
235689	235789	236789	245678
245679	245689	245789	246789
256789	345678	345679	345689
345789	346789	356789	456789

Octahedral type

012678	012679	012689	012789
013678	013679	013689	013789
016789	024678	024679	024689
024789	026789	034678	034679
034689	034789	036789	046789
125678	125679	125689	125789
126789	135678	135679	135689
135789	136789	156789	245678
245679	245689	245789	246789
256789	345678	345679	345689
345789	346789	356789	456789

References

- [CJT26] James Cruickshank, Bill Jackson, and Shin-ichi Tanigawa. Volume rigidity of simplicial manifolds. *Combinatorica*, 46:23, 2026.